Euclid’s extra prime

Euclid IX.20, VII definitions 1–2 and selected clauses of VII.31, in Heiberg’s 1884 Greek edition and nineteenth-century Latin translation. English, numerical examples, division ledger, trial search and descent controls are new teaching work. The bounded computation illustrates cases; the argument for any given finite list is a separate reason.

The chosen list and its added unit

Given distinct primes: 2, 3, 5, 7, 11, 13.

M · common multiple

30030

$2 \times 3 \times 5 \times 7 \times 11 \times 13$

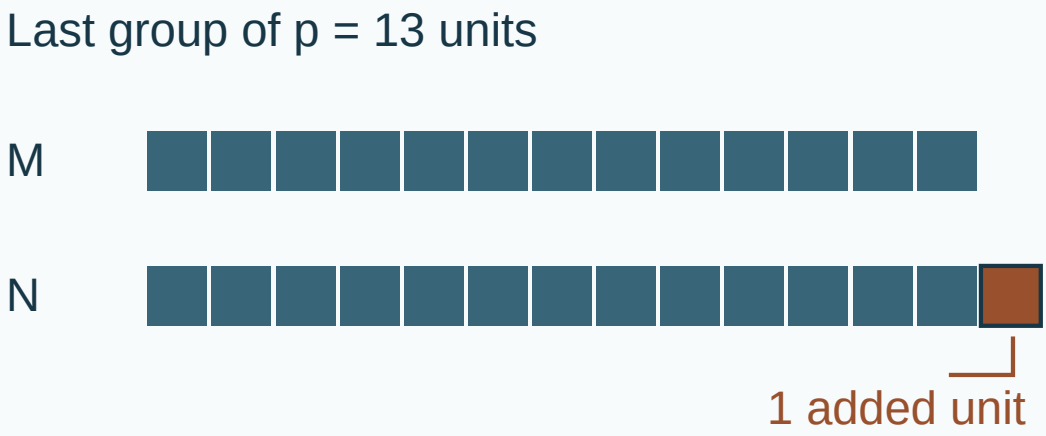
N · one unit added

30031

$30030 + 1$

Each given prime divides M exactly. Each leaves one unit in N.

Given p	Whole copies in M	N ÷ p remainder
p = 2	15015	**1**
p = 3	10010	**1**
p = 5	6006	**1**
p = 7	4290	**1**
p = 11	2730	**1**
p = 13	2310	**1**



Last complete group: 13 equal units. 2309 earlier complete groups omitted. The outlined unit is added to N.

$$30030 = 13 \times 2310$$

$$30031 = 13 \times 2310 + 1$$

The same quotient leaves one unit. Since $p \geq 2$, that remainder is smaller than p and is not zero.

What the bounded search established

Prime trial divisors permitted through 100. Whole-number calculations use exact integers; no probable-prime test is used.

The sum is composite

$30031 = 59 \times 509$

$61^2 > 509$; every smaller prime trial divisor has been excluded or removed.

17 prime trial divisors actually tried: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59. The factorization is complete.

If N is prime

N itself is a prime outside the given list. None of the given primes divides it.

This branch is part of the proof even when this example does not take it.

> ### If N is composite
>
> VII.31 guarantees a prime H dividing N. If H were given, it would divide M too, and then N − M = 1. A prime at least two cannot divide one.
>
> This example supplies H = 59. It divides 30031, and is absent from the list.

A reason for any finite list

1. Take any nonempty finite list of distinct primes. Their product M is a positive whole number divisible by every listed prime.
2. N = M + 1 is greater than one. For each given prime p, N leaves remainder 1 when divided by p. Hence no given prime divides N.
3. If N is prime, take H = N. Otherwise VII.31 supplies a prime divisor H of N.
4. In either case H is a prime outside the given list. The given finite list therefore cannot contain every prime.

The product need not be prime; H need not be the next prime or greater than every given prime. The argument works for an arbitrary finite list, independent of the twelve-prime input limit or the 100,000 trial bound.

For distinct primes the least common multiple equals the product: in any common multiple, divisibility by one prime and Euclid’s lemma (VII.30) retain the other distinct prime factors after division. Repeating this shows their product divides every common multiple. The product is itself a common multiple, so is least. The bench uses that equality; the contradiction needs only a common multiple.

A separate VII.31 descent

This small example starts at 2310, independently of the chosen prime list. Every step chooses a smaller divisor greater than one; each still divides the original.

2310
2310 = 2310 × 1

1155
2310 = 1155 × 2; 2310 = 1155 × 2

385
1155 = 385 × 3; 2310 = 385 × 6

77
385 = 77 × 5; 2310 = 77 × 30

11
77 = 11 × 7; 2310 = 11 × 210

> 11 has no proper divisor greater than one: the descent has reached a prime. It still divides 2310.

Only finitely many whole numbers at least two lie below a fixed starting number. An endlessly smaller chain cannot occur. If a chosen divisor is composite there is another proper divisor; eventually a prime must occur. This argument supplies a prime divisor without relying on the app’s trial bound.

Read the argument in three languages

IX.20 · statement and givens

1. More than the given multitude

Greek · p.388

Οἱ πρῶτοι ἀριθμοὶ πλείους εἰσὶ παντὸς τοῦ προτεθέντος πλήθους πρώτων ἀριθμῶν. Ἔστωσαν οἱ προτεθέντες πρῶτοι ἀριθμοὶ οἱ Α, Β, Γ· λέγω, ὅτι τῶν Α, Β, Γ πλείους εἰσὶ πρῶτοι ἀριθμοί.

Heiberg’s Latin · p.389

Primi numeri plures sunt quauis data multitudine primorum numerorum.
Sint dati numeri primi A, B, Γ. dico, plures esse primos numeros quam A, B, Γ.

There are more prime numbers than any given multitude of prime numbers. Let the given primes be A, B, Γ. I say that there are more primes than A, B, Γ.

The comparison is about how many primes, not about making the next prime larger than every given one. The printed proof names three givens. The modern bench accepts up to twelve; the same divisibility reason applies to any nonempty finite list. That extension is explained below, rather than inferred from the app’s finite limit.

> **πλείους / plures**
>
> more in number; not ‘larger in value’. The genitive τῶν Α, Β, Γ supplies the comparison.

IX.20 · construction

2. Take the least common multiple, then a unit

Greek · p.388

Εἰλήφθω γὰρ ὁ ὑπὸ τῶν Α, Β, Γ ἐλάχιστος μετρούμενος καὶ ἔστω ὁ ΔΕ, καὶ προσκείσθω τῷ ΔΕ μονὰς ἡ ΔΖ.

Heiberg’s Latin · p.389

sumatur enim, quem minimum metiuntur A, B, Γ, et sit ΔΕ, et numero ΔΕ adiiciatur unitas ΔΖ.

Take the least number measured by A, B, Γ, and let it be ΔΕ; let the unit ΔΖ be added to ΔΕ.

‘Measured by’ means divisible by, without a remainder. The source takes a least common multiple, citing VII.36 in the Latin edition. For distinct primes this equals their product, our M. Any common multiple would suffice for the contradiction. The product formula, equals sign and letter M are modern aids. On p.389 the printed line orders Δ–Ε–Ζ, although the text names ΔΖ as the unit and ΕΖ as the sum; the bench does not copy that endpoint order.

> **ἐλάχιστος μετρούμενος / quem minimum metiuntur**
>
> the least [number] measured [by them]; minimum is a common multiple, not a new prime.

IX.20 · the two branches

3. Either prime, or not

Greek · p.388 and p.390

ὁ δὴ ΕΖ ἤτοι πρῶτός ἐστιν ἢ οὔ. ἔστω πρότερον πρῶτος· εὑρημένοι ἄρα εἰσὶ πρῶτοι ἀριθμοὶ οἱ Α, Β, Γ, ΕΖ πλείους τῶν Α, Β, Γ.

Heiberg’s Latin · p.389 and p.391

EZ igitur aut primus est aut non primus. prius sit primus. ergo inuenti sunt primi numeri A, B, Γ, EZ plures numeris A, B, Γ.

The number EZ is therefore either prime or not. First suppose it prime: then the primes A, B, Γ, EZ have been found, more than A, B, Γ.

Greek begins on p.388 and continues on p.390; the Latin crosses pp.389–391. This is a case distinction, not a claim that adding a unit always produces a prime. On the bench N = M + 1 is EZ. If N is prime it supplies the additional prime itself.

> ### ἤτοι … ἢ οὔ / aut … aut non
>
> either … or not. Both possibilities belong to the proof. Greek πρῶτος literally has the sense ‘first’, here ‘prime’.

IX.20 · the composite branch

4. Suppose it is not prime

Greek · p.390

Ἀλλὰ δὴ μὴ ἔστω ὁ EZ πρῶτος· ὑπὸ πρώτου ἄρα τινὸς ἀριθμοῦ μετρεῖται. μετρείσθω ὑπὸ πρώτου τοῦ H· λέγω, ὅτι ὁ H οὐδενὶ τῶν A, B, Γ ἐστιν ὁ αὐτός.

Heiberg’s Latin · p.391

uerum ne sit EZ primus. itaque primus aliquis numerus eum metitur. metiatur primus numerus H. dico, numerum H nulli numerorum A, B, Γ aequalem esse.

Now suppose EZ is not prime. It is therefore measured by some prime number. Let the prime H measure it. I say that H is the same as none of A, B, Γ.

VII.31 supplies the existence of a prime divisor. It does not depend on the trial search in this app. For the opening six-prime list, 30031 = 59 × 509. Either proven prime factor supplies H. The sum itself need not be prime, and you do not need a complete factorization to obtain one prime divisor.

> **μετρεῖται / μετρείσθω / metitur / metiatur**
>
> is measured / let it be measured. The Greek passive construction puts the measured number first; Latin makes the prime the measuring subject.

IX.20 · temporary assumption

5. Assume that divisor was already given

Greek · p.390

εἰ γὰρ δυνατόν, ἔστω. οἱ δὲ A, B, Γ τὸν ΔΕ μετροῦσιν· καὶ ὁ Η ἄρα τὸν ΔΕ μετρήσει. μετρεῖ δὲ καὶ τὸν ΕΖ·

Heiberg's Latin · p.391

nam si fieri potest, sit. uerum A, B, Γ numerum ΔΕ metiuntur. itaque etiam H numerum ΔΕ metitur. uerum etiam numerum EZ metitur.

For if possible, let it be so. A, B, Γ measure ΔΕ; H will therefore measure ΔΕ too. But it also measures EZ.

This is a temporary supposition: H equals one of the given primes. It is not an extra condition placed on H. In modern symbols H would divide both M and N. Subtracting two whole multiples of H leaves another whole multiple of H.

> **εἰ … δυνατόν / si fieri potest**
>
> if it is possible; introduces the assumption that will be rejected.

IX.20 · contradiction

6. Then a number would measure the unit

Greek · p.390

καὶ λοιπὴν τὴν ΔΖ μονάδα μετρήσει ὁ Η ἀριθμὸς ὤν· ὅπερ ἄτοπον. οὐκ ἄρα ὁ Η ἐνὶ τῶν A, B, Γ ἐστιν ὁ αὐτός. καὶ ὑπόκειται πρῶτος.

Heiberg's Latin · p.391

quare etiam quae relinquitur, unitatem ΔZ metietur H, qui numerus est; quod absurdum est. ergo H nulli numerum A, B, Γ aequalis est. et suppositum est, H primum esse.

It would therefore also measure the remaining unit ΔZ, though H is a number: which is impossible. H is therefore the same as none of A, B, Γ, and it was assumed prime.

In this arithmetic, a number is a multitude of units (VII definition 2), distinct from the unit. H is at least two, so cannot divide one into a whole number of copies. Compare the exact remainder 1 in every ledger row. Heiberg's printed Latin has 'nulli numerum' here; this selection retains that wording rather than silently rewriting it as 'nulli numerorum'.

> ἀριθμὸς ὤν / qui numerus est
>
> being a number; the distinction between number and unit is doing mathematical work.

IX.20 · conclusion

7. An additional prime has been found

Greek · p.390

εὑρημένοι ἄρα εἰσὶ πρῶτοι ἀριθμοὶ πλείους τοῦ προτεθέντος πλήθους τῶν A, B, Γ οἱ A, B, Γ, H· ὅπερ ἔδει δεῖξαι.

Heiberg's Latin · p.391

ergo inuenti sunt primi numeri A, B, Γ, H plures data multitudine A, B, Γ; quod erat demonstrandum.

The primes A, B, Γ, H have therefore been found, more than the given multitude A, B, Γ: which was to be shown.

Neither branch needs H to be the next prime in numerical order. With givens 3, 5, 7 the sum 106 has prime divisor 2, smaller than all three givens. A finite list cannot be the complete collection: the argument always supplies an additional prime outside it. Repeating a few numerical examples illustrates this reason but does not prove it by enumeration.

> **εὑρημένοι / inuenti**
>
> having been found; a completed result in either branch, without a promise of a fast factoring algorithm.

VII · definition 1

8. The unit

Greek · p.184

Μονάς ἐστιν, καθ’ ἣν ἕκαστον τῶν ὄντων ἓν λέγεται.

Heiberg’s Latin · p.185

Unitas est ea, secundum quam unaquaeque res una nominatur.

A unit is that by virtue of which each existing thing is called one.

This is Euclid’s definition of a unit, not a modern set-theoretic definition of 1. The bench writes the added unit as 1, while retaining the historical distinction needed in the contradiction.

> **μονάς / μονάδα / unitas / unitatem**
>
> unit: nominative names it, accusative is used as the object of ‘will measure’ in IX.20.

VII · definition 2

9. A number is a multitude of units

Greek · p.184

Ἀριθμὸς δὲ τὸ ἐκ μονάδων συγκείμενον πλῆθος.

Heiberg’s Latin · p.185

Numerus autem est multitudo ex unitatibus composita.

A number is a multitude composed of units.

Do not read this as a claim about every modern use of the word number. The source distinguishes the unit from a multitude. The given-prime controls require integers at least two; accepting 1 as a prime would destroy the ‘none of the givens divides N’ step, since 1 divides everything.

> ### ἐκ μονάδων / ex unitatibus
> out of units; Greek ἐκ takes the genitive, Latin ex the ablative.

VII.31 · statement

10. Where does a prime divisor come from?

Greek · p.250

Ἅπας σύνθετος ἀριθμὸς ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται.

Heiberg’s Latin · p.251

Quemuis numerum compositum primus aliquis numerus metitur.

Every composite number is measured by some prime number.

The full printed proof is on pp.250–253. It repeatedly takes a divisor of a composite divisor; a divisor of a divisor also divides the original. The separate 2310 descent lets you choose such proper divisors. It does not require you to know which of them is prime before choosing.

> ### σύνθετος / compositus
> composite, put together; the divisor chain eventually reaches a prime.

VII.31 · selected impossibility clause

11. A descent cannot continue forever

Greek · p.252

τὸν Α ἀριθμὸν ἄπειροι ἀριθμοί, ὧν ἕτερος ἑτέρου ἐλάσσων ἐστίν· ὅπερ ἐστὶν ἀδύνατον ἐν ἀριθμοῖς.

Heiberg’s Latin · p.253

infiniti numeri numerum A metientur, alter semper altero deinceps minores; quod in numeris fieri non potest.

Infinitely many numbers would measure A, each successively smaller than another: which is impossible among numbers.

The Greek selection starts in the middle of the sentence continuing from p.250; its introductory ‘would measure’ is there. This is a strict descent through whole numbers at least two. Below a fixed starting number only finitely many such numbers exist. If every chosen divisor were composite, you could descend again, so eventually one must be prime. This is an existence argument, not an assertion that our chosen search bound will find it.

> ### ἐλάσσων / minores
>
> smaller; unlike πλείους in IX.20, this comparison is about numerical size.

Retained settings and native files


```
{
  "format": "stoicheia-primes",
  "version": 1,
  "settings": {
    "primes": [
      "2",
      "3",
      "5",
      "7",
      "11",
      "13"
    ],
    "limit": 100,
    "selected": 5,
    "reading": 5,
    "page": 390,
    "descent": [
      2310,
      1155,
      385,
      77,
      11
    ]
  }
}
```

The CSV retains this calculation and the separate descent chain, not a log of learner actions. Reopen these settings in the workshop. The link needs the app; this reading’s text, diagram, pages, font and native downloads work from disk without scripts or network assets.

The complete printed evidence

J. L. Heiberg, *Euclidis Elementa*, volume II, Leipzig: B. G. Teubner, 1884. The title prints MDCCCLXXXIV. IX.20 is on pp.388–391; VII definitions 1–2 on pp.184–185; VII.31 on pp.250–253. Whole pages retain margins, editorial apparatus, neighbouring propositions and original markings. Numbered print pages plus 26 give the one-based PDF page; title is PDF p.7.

Selections join line breaks and line-end hyphens, omit superscript reference markers, normalize printed theta glyphs and lightly regularize punctuation. Greek and Latin letter labels are rendered in Greek Unicode; Latin u/v spellings remain. IX.20 is transcribed in full across seven reading units. VII definitions 1–2 are complete; VII.31 is represented by its statement and a selected impossibility clause, with the complete printed proof retained in the images. The p.389 diagram’s endpoint order differs from the text’s names for the added unit and sum, as explained in reading 2. New English and contextual language notes have not had independent philological review.

Original public-domain volume. PDF SHA-256: cd24f03c6e4e6d5f30a908d144c8f0f7003e99813334edb3a4107d748cb41043. Derivatives: whole pages, 2200 px long edge, JPEG quality 88.

νατόν ἐστι τοῖς *A*, *B*, *Γ* τέταρτον ἀνάλογον προσευρεῖν ἀριθμόν. εἰ γὰρ δυνατόν, προσευρήσθω ὁ *E*· ὁ ἄρα ἐκ τῶν *A*, *E* ἴσος ἐστὶ τῷ ἐκ τῶν *B*, *Γ*. ἀλλὰ ὁ ἐκ τῶν *B*, *Γ* ἐστιν ὁ *Δ*· καὶ ὁ ἐκ τῶν *A*, *E* ἄρα ἴσος ἐστὶ τῷ *Δ*. ὁ *A* ἄρα τὸν *E* πολλαπλασιάσας τὸν *Δ* πεποίηκεν· ὁ *A* ἄρα τὸν *Δ* μετρεῖ κατὰ τὸν *E*· ὥστε μετρεῖ ὁ *A* τὸν *Δ*. ἀλλὰ καὶ οὐ μετρεῖ· ὅπερ ἄτοπον. οὐκ ἄρα δυνατόν ἐστι τοῖς *A*, *B*, *Γ* τέταρτον ἀνάλογον προσευρεῖν ἀριθμόν, ὅταν ὁ *A* τὸν *Δ* μὴ μετρῇ. ἀλλὰ δὴ οἱ *A*, *B*, *Γ* μήτε ἑξῆς ἔστωσαν ἀνάλογον μήτε οἱ ἄκροι πρῶτοι πρὸς ἀλλήλους. καὶ ὁ *B* τὸν *Γ* πολλαπλασιάσας τὸν *Δ* ποιείτω. ὁμοίως δὴ δειχθήσεται, ὅτι εἰ μὲν μετρεῖ ὁ *A* τὸν *Δ*, δυνατόν ἐστιν αὐτοῖς ἀνάλογον προσευρεῖν, εἰ δὲ οὐ μετρεῖ, ἀδύνατον· ὅπερ ἔδει δεῖξαι.

κ'.

Οἱ πρῶτοι ἀριθμοὶ πλείους εἰσὶ παντὸς τοῦ προτεθέντος πλήθους πρώτων ἀριθμῶν.

Ἔστωσαν οἱ προτεθέντες πρῶτοι ἀριθμοὶ οἱ *A*, *B*, *Γ*· λέγω, ὅτι τῶν *A*, *B*, *Γ* πλείους εἰσὶ πρῶτοι ἀριθμοί.

Εἰλήφθω γὰρ ὁ ὑπὸ τῶν *A*, *B*, *Γ* ἐλάχιστος μετρούμενος καὶ ἔστω ὁ *ΔE*, καὶ προσκείσθω τῷ *ΔE* μονὰς ἡ *ΔZ*. ὁ δὴ *EZ* ἤτοι πρῶτός ἐστιν ἢ οὔ.

---

1. ἐστιν P. 2. προσηυρήσθω FV. 3. ἀλλ' BFV. 10. μή] supra m. 1 F, οὐ supra m. 2 V. μετρήσῃ F, μετρεῖ q. ἀλλὰ δή — 15: ἀδύνατον] om. BVq. 10. δή] μήτε F. ἑξῆς] οἱ ἑξῆς F. 12. ποιήτω φ (non F). 14. αὐτοῖς] αὐτοῖς τετάρτοις F. εἰ δέ] οὐδ' F. 15. ὅπερ ἔδει δεῖξαι] om. Bq. 17. πρῶτοι ἀριθμοί] del. et supra scr. πρώτων ἀριθμῶν m. 2 B. 18. προστεθέντος F. 23. καί] m. 2 B, om. V. 24. ΔZ] AZ F.

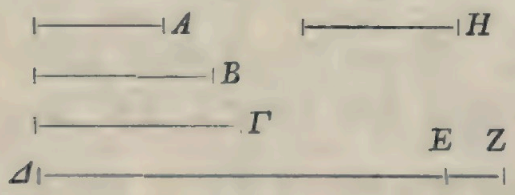
Printed p.388 · PDF p.414 · SHA-256 452a63259825c59bd9bcdc1f6a7179d9eda5c0d68cbd6ee7be05955d4dbc7cc2

numerorum A, B, Γ quartum proportionalem inueniri non posse. nam si fieri potest, inueniatur E. itaque $A \times E = B \times \Gamma$ [VII, 19]. uerum $B \times \Gamma = \Delta$. quare $A \times E = \Delta$. itaque A numerum E multiplicans numerum Δ effecit. A igitur numerum Δ secundum E metitur. itaque A numerum Δ metitur. uerum etiam non metitur; quod absurdum est. ergo numerorum A, B, Γ quartus proportionalis inueniri non potest, ubi A numerum Δ non metitur. uerum A, B, Γ ne sint deinceps proportionales neu extremi inter se primi. et sit $B \times \Gamma = \Delta$. similiter demonstrabimus, si A numerum Δ metiatur, fieri posse, ut eorum quartus[1]) inueniatur proportionalis, sin non metiatur, fieri non posse; quod erat demonstrandum.

XX.

Primi numeri plures sunt quauis data multitudine primorum numerorum.

Sint dati numeri primi A, B, Γ. dico, plures esse primos numeros quam A, B, Γ. sumatur enim, quem



minimum metiuntur A, B, Γ [VII, 36] et sit ΔE, et numero ΔE adiiciatur unitas ΔZ. EZ igitur aut primus est aut non primus. prius sit primus. ergo in-

1) Uidetur scribendum esse lin. 14: *αὐτοῖς τέταρτον ἀνάλογον*; cfr. F.

Printed p.389 · PDF p.415 · SHA-256 0a11f4afa024dd836be31f98ae7cefea60ec9bd90ecaab3008c2b9ae59191cb9

ἔστω πρότερον πρῶτος· εὑρημένοι ἄρα εἰσὶ πρῶτοι ἀριθμοὶ οἱ *A*, *B*, *Γ*, *EZ* πλείους τῶν *A*, *B*, *Γ*.

Ἀλλὰ δὴ μὴ ἔστω ὁ *EZ* πρῶτος· ὑπὸ πρώτου ἄρα τινὸς ἀριθμοῦ μετρεῖται. μετρείσθω ὑπὸ πρώτου τοῦ *H*· λέγω, ὅτι ὁ *H* οὐδενὶ τῶν *A*, *B*, *Γ* ἐστιν ὁ αὐτός. εἰ γὰρ δυνατόν, ἔστω. οἱ δὲ *A*, *B*, *Γ* τὸν *ΔE* μετροῦσιν· καὶ ὁ *H* ἄρα τὸν *ΔE* μετρήσει. μετρεῖ δὲ καὶ τὸν *EZ*· καὶ λοιπὴν τὴν *ΔZ* μονάδα μετρήσει ὁ *H* ἀριθμὸς ὤν· ὅπερ ἄτοπον. οὐκ ἄρα ὁ *H* ἑνὶ τῶν *A*, *B*, *Γ* ἐστιν ὁ αὐτός. καὶ ὑπόκειται πρῶτος. εὑρημένοι ἄρα εἰσὶ πρῶτοι ἀριθμοὶ πλείους τοῦ προτεθέντος πλήθους τῶν *A*, *B*, *Γ* οἱ *A*, *B*, *Γ*, *H*· ὅπερ ἔδει δεῖξαι.

κα'.

Ἐὰν ἄρτιοι ἀριθμοὶ ὁποσοιοῦν συντεθῶσιν, ὁ ὅλος ἄρτιός ἐστιν.

Συγκείσθωσαν γὰρ ἄρτιοι ἀριθμοὶ ὁποσοιοῦν οἱ *AB*, *BΓ*, *ΓΔ*, *ΔE*· λέγω, ὅτι ὅλος ὁ *AE* ἄρτιός ἐστιν.

Ἐπεὶ γὰρ ἕκαστος τῶν *AB*, *BΓ*, *ΓΔ*, *ΔE* ἄρτιός ἐστιν, ἔχει μέρος ἥμισυ· ὥστε καὶ ὅλος ὁ *AE* ἔχει μέρος ἥμισυ. ἄρτιος δὲ ἀριθμός ἐστιν ὁ δίχα διαιρούμενος· ἄρτιος ἄρα ἐστὶν ὁ *AE*· ὅπερ ἔδει δεῖξαι.

---

1. Supra *πρῶτος* add. *ἡ EZ* m. rec. V. *εἰσίν* P, *εἰσὶν οἱ* q. 2. *ἀριθμοί*] om. F. *Γ*] (prius) *ΓΔ* F, *Δ* del. m. 1. 6. *δυνατόν, ἔστω*] *ὁ H ἑνὶ τῶν A, B, Γ ἐστιν ὁ αὐτός* Theon (BFVq). 7. *ΔE*] ZE F. *μετροῦσι* BFVq. *ΔE*] ZE F. 8. *καί*] *καὶ ὁ H* F. *EZ*] *ΔE* F. 10. *καί*] *ὁ αὐτὸς δὲ καί* P. 11. *εἰσί*] *εἰσὶν οἱ* V. 13. *H*] *H ἄρα* ante ras. 6 litt. F. 15. *συν* — supra scr. B. 16. *ἐστί* Vq, comp. F. 17. *ὁποσοιοῦν*] e corr. V. 18. *BΓ*] in ras. P. *ΓΔ*] m. 2 V. 21. *καί*] supra lac. pergam. m. rec. F. 23. *ὁ AE ἄρα ἐστίν* F. *ὅπερ ἔδει δεῖξαι*] om. B.

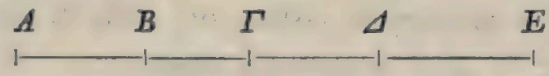
Printed p.390 · PDF p.416 · SHA-256 c745b20db69c37d4d4dfb1a6c90a4a8bea6ac29f447d7c5493c8075486981b30

uenti sunt primi numeri A, B, Γ, EZ plures numeris A, B, Γ. uerum ne sit EZ primus. itaque primus aliquis numerus eum metitur [VII, 31]. metiatur primus numerus H. dico, numerum H nulli numerorum A, B, Γ aequalem esse. nam si fieri potest, sit. uerum A, B, Γ numerum ΔE metiuntur. itaque etiam H numerum ΔE metitur. uerum etiam numerum EZ metitur. quare etiam[1]) quae relinquitur, unitatem ΔZ metietur H, qui numerus est; quod absurdum est. ergo H nulli numerum A, B, Γ aequalis est. et suppositum est, H primum esse. ergo inuenti sunt primi numeri A, B, Γ, H plures data multitudine A, B, Γ; quod erat demonstrandum.

XXI.

Si quotlibet numeri pares componuntur, totus par erit.

Componantur enim quotlibet numeri pares AB, $B\Gamma$, $\Gamma\Delta$, ΔE. dico, etiam totum AE parem esse.



nam quoniam singuli numeri AB, $B\Gamma$, $\Gamma\Delta$, ΔE pares sunt, partem dimidiam habent [VII def. 6]. quare etiam totus AE partem dimidiam habet. par autem numerus is est, qui in duas partes aequales diuiditur [id.]. ergo AE par est; quod erat demonstrandum.

1) U. ad VII, 28.

Printed p.391 · PDF p.417 · SHA-256 4273d570062256e0c22643e92e161fa6b5779d5ed026af912a41ecad784ff15c

ζ'.

Ὅροι.

α'. Μονάς ἐστιν, καθ' ἣν ἕκαστον τῶν ὄντων ἓν λέγεται.

β'. Ἀριθμὶς δὲ τὸ ἐκ μονάδων συγκείμενον πλῆθος.

γ'. Μέρος ἐστὶν ἀριθμὸς ἀριθμοῦ ὁ ἐλάσσων τοῦ μείζονος, ὅταν καταμετρῇ τὸν μείζονα.

δ'. Μέρη δέ, ὅταν μὴ καταμετρῇ.

ε'. Πολλαπλάσιος δὲ ὁ μείζων τοῦ ἐλάσσονος, ὅταν καταμετρῆται ὑπὸ τοῦ ἐλάσσονος.

ς'. Ἄρτιος ἀριθμός ἐστιν ὁ δίχα διαιρούμενος.

ζ'. Περισσὸς δὲ ὁ μὴ διαιρούμενος δίχα ἢ [ὁ] μονάδι διαφέρων ἀρτίου ἀριθμοῦ.

η'. Ἀρτιάκις ἄρτιος ἀριθμός ἐστιν ὁ ὑπὸ ἀρτίου ἀριθμοῦ μετρούμενος κατὰ ἄρτιον ἀριθμόν.

θ'. Ἀρτιάκις δὲ περισσός ἐστιν ὁ ὑπὸ ἀρτίου ἀριθμοῦ μετρούμενος κατὰ περισσὸν ἀριθμόν.

Def. 3—5: Psellus p. 7. 6—7: Martianus Capella VII, 748. 8. Iamblichus in Nicom. p. 27. Philop. in Nicom. ed. Hoche 1864 p. 16. 9. Iamblichus p. 31.

---

1. ὅροι] om. PB. numeros om. codd. 2. ἐστι PBFp. ἣν] ὅ BFV. 10. ἐλάττονος V. 12. ὁ] om. P. 14. προσυπακουστέον· μόνον P mg. m. 1. 16. ἐστιν] ἀριθμός ἐστιν P, ἐστιν ἀριθμός p. κἀνταῦθα προσυπακουστέον· μόνον mg. m. 1 P. τοῦ ἀρτίου deleto τοῦ V.

Printed p.184 · PDF p.210 · SHA-256 8727e5fad54f649a9c720d3faf9a03d455dd500098a0d3d97832b9668910f5df

VII.

Definitiones.

1. Unitas est ea, secundum quam unaquaeque res una nominatur.

2. Numerus autem est multitudo ex unitatibus composita.

3. Pars est minor numerus maioris, ubi maiorem metitur.

4. Partes autem, ubi non metitur.

5. Multiplex autem maior minoris, ubi minor eum metitur.

6. Par numerus est, qui in duas partes aequales diuiditur.

7. Impar autem, qui in duas partes aequales non diuiditur, siue qui unitate differt a pari numero.

8. Pariter par est numerus, quem par numerus secundum parem numerum metitur.[1])

9. Pariter autem impar est, quem par numerus secundum imparem numerum metitur.[2])

---

1) Def. 8 scriptor nescio quis, qui Philoponi commentarium in Nicomachum retractauit, apud Hoche Philop. 1865 p. V in quibusdam *ἀντιγράφοις* ita inuenit expressam: *ἀρτιάκις ἄρτιός ἐστιν ἀριθμὸς ὁ ὑπὸ ἀρτίου ἀριθμοῦ κατὰ ἄρτιον ἀριθμὸν μόνως μετρούμενος*, de qua scriptura falsa u. Studien p. 200.

2) De def. *ι'* interpolata u. Studien p. 198 sq.; om. ed Basil. et Gregorius.

Printed p.185 · PDF p.211 · SHA-256 0839aca8cdfadb90781a21f885cc15dfead2fc3174e497efda617ab502115ce2

τῶν *A*, *B*. ἔστιν ἄρα ὡς ὁ *Δ* πρὸς τὸν *A*, οὕτως ὁ *B* πρὸς τὸν *E*. οἱ δὲ *Δ*, *A* πρῶτοι, οἱ δὲ πρῶτοι καὶ ἐλάχιστοι, οἱ δὲ ἐλάχιστοι μετροῦσι τοὺς τὸν αὐτὸν λόγον ἔχοντας ἰσάκις ὅ τε μείζων τὸν μείζονα καὶ ὁ ἐλάσσων τὸν ἐλάσσονα, τουτέστιν ὅ τε ἡγούμενος τὸν ἡγούμενον καὶ ὁ ἑπόμενος τὸν ἑπόμενον· ὁ *Δ* ἄρα τὸν *B* μετρεῖ. ὁμοίως δὴ δείξομεν, ὅτι καὶ ἐὰν τὸν *B* μὴ μετρῇ, τὸν *A* μετρήσει. ὁ *Δ* ἄρα ἕνα τῶν *A*, *B* μετρεῖ· ὅπερ ἔδει δεῖξαι.

λα'.

Ἅπας σύνθετος ἀριθμὸς ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται.

Ἔστω σύνθετος ἀριθμὸς ὁ *A*· λέγω, ὅτι ὁ *A* ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται.

Ἐπεὶ γὰρ σύνθετός ἐστιν ὁ *A*, μετρήσει τις αὐτὸν ἀριθμός. μετρείτω, καὶ ἔστω ὁ *B*. καὶ εἰ μὲν πρῶτός ἐστιν ὁ *B*, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν ἀριθμός. μετρείτω, καὶ ἔστω ὁ *Γ*. καὶ ἐπεὶ ὁ *Γ* τὸν *B* μετρεῖ, ὁ δὲ *B* τὸν *A* μετρεῖ, καὶ ὁ *Γ* ἄρα τὸν *A* μετρεῖ. καὶ εἰ μὲν πρῶτός ἐστιν ὁ *Γ*, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν ἀριθμός. τοιαύτης δὴ γινομένης ἐπισκέψεως ληφθήσεταί τις πρῶτος ἀριθμός, ὃς μετρήσει. εἰ γὰρ οὐ ληφθήσεται, μετρήσουσι

---

3. καί] om. p. οἱ δὲ ἐλάχιστοι] supra P, in mg. m. 1 Vφ. 4. μείζων τόν] mg. φ (τόν in ras.). 6. τόν] (prius) in ras. φ. 8. *B* μή] in ras. p; μή supra V m. 2. Post μετρῇ ras. 1 litt. p. 9. Sequitur in BVpφ alia demonstratio prop. XXXI a Theone addita; u. app. 10. λγ' BVφ, P m. rec.; λδ' p. 14. μετρῆται P, corr. m. 2. 17. δῆλον ἂν εἴη τὸ ζητούμενον Theon (BVpφ). 20. μετρεῖ] (prius)

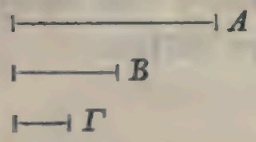
Printed p.250 · PDF p.276 · SHA-256 29ca2948a167bea2ae1b63815c6ae46401c531bf720c3328c2d7a5c28638497e

[prop. XIX] $\Delta : A = B : E$. uerum Δ, A primi sunt, primi autem etiam minimi sunt [prop. XXI], minimi autem eos, qui eandem rationem habent, aequaliter metiuntur, maior maiorem et minor minorem [prop. XX], h. e. praecedens praecedentem et sequens sequentem. itaque Δ numerum B metitur. similiter demonstrabimus, Δ numerum, si B numerum non metiatur, numerum A metiri. ergo Δ alterutrum numerorum A, B metitur; quod erat demonstrandum.

XXXI.

Quemuis numerum compositum primus aliquis numerus metitur.

Sit numerus compositus A. dico, primum aliquem numerum numerum A metiri.



nam quoniam A compositus est, numerus aliquis eum metietur. metiatur et sit B. et si B primus est, factum erit id, quod iussi sumus.[1]) sin compositus est, numerus aliquis eum metietur. metiatur et sit Γ. et quoniam Γ numerum B metitur, et B numerum A metitur, etiam Γ numerum A metitur. et si Γ primus est, factum erit, quod iussi sumus; sin compositus,

1) Sc. primum numerum numerum A metientem inuenire. quamquam haec formula in problemata magis cadit, id quod magis etiam de p. 252, 12 ualet.

om. p. 21. *δῆλον ἂν εἴη τὸ ζητούμενον* Theon (BVpφ). 22. Post *ἀριθμός* add. p: *μετρείτω καὶ ἔστω ὁ Γ. καὶ ἐπεὶ ὁ Γ τὸν B μετρεῖ ὁ δὲ B τὸν A μετρεῖ, καὶ ὁ Γ ἄρα τὸν A μετρεῖ.* 23. *δή*] corr. ex *δέ* V, *δέ* pφ. 24. *ὅς*] supra m. 2 P. Post *μετρήσει* add. Theon *τὸν πρὸ ἑαυτοῦ*, (huc usque etiam P mg. m. rec.) *ὃς καὶ τὸν A μετρήσει* (BVpφ). *εἰ*] corr. ex *ἡ* m. rec. P. *οὐ*] *μή* August.

Printed p.251 · PDF p.277 · SHA-256 b4aa371ff5a00b59a753669474152fcaa24a3097107a1e1964d00930496d7d59

τὸν *A* ἀριθμὸν ἄπειροι ἀριθμοί, ὧν ἕτερος ἑτέρου ἐλάσσων ἐστίν· ὅπερ ἐστὶν ἀδύνατον ἐν ἀριθμοῖς. ληφθήσεταί τις ἄρα πρῶτος ἀριθμός, ὃς μετρήσει τὸν πρὸ ἑαυτοῦ, ὃς καὶ τὸν *A* μετρήσει.

Ἅπας ἄρα σύνθετος ἀριθμὸς ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται· ὅπερ ἔδει δεῖξαι.

λβ'.

Ἅπας ἀριθμὸς ἤτοι πρῶτός ἐστιν ἢ ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται.

Ἔστω ἀριθμὸς ὁ *A*· λέγω, ὅτι ὁ *A* ἤτοι πρῶτός ἐστιν ἢ ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται.

Εἰ μὲν οὖν πρῶτός ἐστιν ὁ *A*, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν. εἰ δὲ σύνθετος, μετρήσει τις αὐτὸν πρῶτος ἀριθμός.

Ἅπας ἄρα ἀριθμὸς ἤτοι πρῶτός ἐστιν ἢ ὑπὸ πρώτου τινὸς ἀριθμοῦ μετρεῖται· ὅπερ ἔδει δεῖξαι.

λγ'.

Ἀριθμῶν δοθέντων ὁποσωνοῦν εὑρεῖν τοὺς ἐλαχίστους τῶν τὸν αὐτὸν λόγον ἐχόντων αὐτοῖς.

Ἔστωσαν οἱ δοθέντες ὁποσοιοῦν ἀριθμοὶ οἱ *A*, *B*, *Γ*· δεῖ δὴ εὑρεῖν τοὺς ἐλαχίστους τῶν τὸν αὐτὸν λόγον ἐχόντων τοῖς *A*, *B*, *Γ*.

Οἱ *A*, *B*, *Γ* γὰρ ἤτοι πρῶτοι πρὸς ἀλλήλους εἰσὶν

---

1. ὁ ἕτερος V φ. τοῦ ἑτέρου BVp φ. 2. ἐστίν] (prius) om. B. 3. πρῶτος ἀριθμός] supra m. 2 V, ἀριθμὸς πρῶτος p. 7. λδ' BV, P m. rec.; λε' p. 8. πᾶς P. 11. ἐστι V φ. 12. γεγονός] Pp, δῆλον BV φ. 13. ἐπιταχθέν] ζητούμενον Theon (BVp φ). 17. λε' BV, P m. rec.; λϛ' p. 19. τοὺς αὐτοὺς λόγους Bp. 22. τοὺς αὐτοὺς λόγους BVp φ.

Printed p.252 · PDF p.278 · SHA-256 cf11c500e29185f5474e44bfd213d4ad10d81555984c003228a239105e20fe75

numerus aliquis eum metietur. hac igitur ratiocinatione procedente inuenietur primus aliquis numerus, qui metietur.[1]) nam si non inuenietur, infiniti numeri numerum *A* metientur, alter semper altero deinceps minores; quod in numeris fieri non potest. itaque inuenietur primus aliquis numerus proxime antecedentem metiens, qui etiam numerum *A* metiatur.

Ergo quemuis numerum compositum primus aliquis numerus metitur; quod erat demonstrandum.

XXXII.

Quiuis numerus aut primus est, aut primus numerus eum metitur.

Sit numerus *A*. dico, numerum *A* aut primum esse aut primum aliquem numerum eum metiri.

A iam si primus est *A*, factum erit, quod iussi sumus; sin compositus, primus aliquis numerus eum metietur [prop. XXXI].

Ergo quiuis numerus aut primus est, aut primus numerus eum metitur; quod erat demonstrandum.

XXXIII.

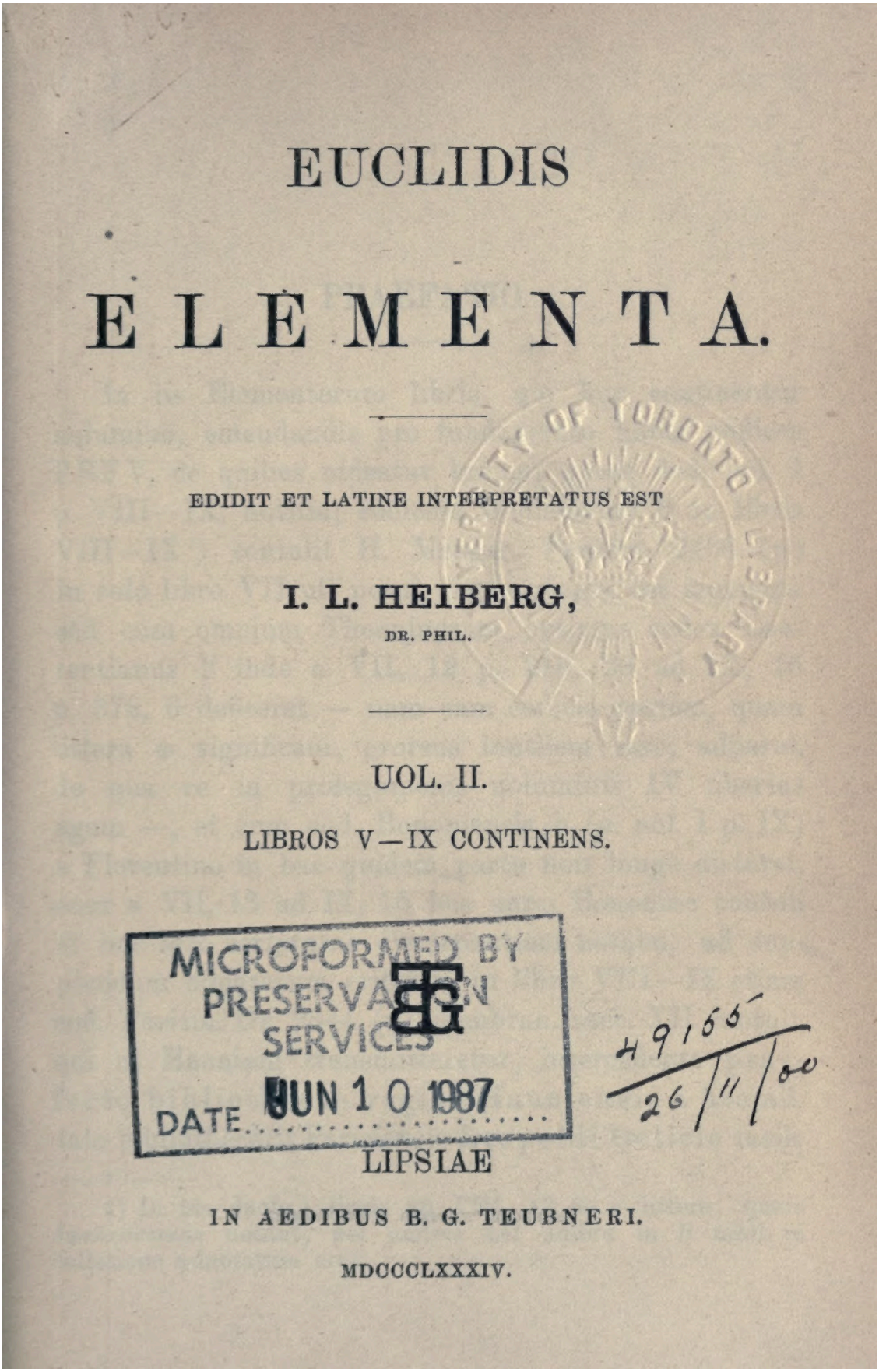
Datis quotlibet numeris minimos eorum, qui eandem rationem habent, inuenire.

Dati sint quotlibet numeri *A*, *B*, *Γ*. oportet igitur minimos eorum inuenire, qui eandem rationem habeant ac *A*, *B*, *Γ*.

A, *B*, *Γ* enim aut inter se primi sunt aut non

1) Sc. numerum praecedentem et ea de causa numerum *A* (cfr. lin. 4). et puto, haec audiri posse. etsi fieri potest, ut haec uerba in P mero errore ob ὁμοιοτέλευτον exciderint.

Printed p.253 · PDF p.279 · SHA-256 3cef46673bea2b9cce020800ec7c127e78f8fdd1a7a8bdec6d4f1a84d95fc14d



EUCLIDIS

ELEMENTA.

EDIDIT ET LATINE INTERPRETATUS EST

I. L. HEIBERG,

DR. PHIL.

UOL. II.

LIBROS V—IX CONTINENS.

MICROFORMED BY PRESERVATION SERVICES
DATE JUN 10 1987
49165
26/11/00

LIPSIAE

IN AEDIBUS B. G. TEUBNERI.

MDCCCLXXXIV.

Title · PDF p.7 · SHA-256 f1bcf7e5c1dd45e498112668d3dde1025224e4587619e47e37ba633617c110e1

Licenses and limits

Euclid IX.20, VII definitions 1–2 and selected clauses of VII.31, in Heiberg’s 1884 Greek edition and nineteenth-century Latin translation. English, numerical examples, division ledger, trial search and descent controls are new teaching work. The bounded computation illustrates cases; the argument for any given finite list is a separate reason.

New work is MIT licensed. Historical text and source scans are public domain. Bundled Noto Serif is under SIL OFL 1.1. Input is restricted to twelve distinct proven primes at most 10,000,000,000; trial search stops at the chosen bound, at most 100,000. The separate descent starts at 2310. Those computational limits do not bound the theorem. No formal proof assistant, physical print or comprehensive assistive-technology review is claimed.

▼ Complete MIT and font licenses


```
MIT License

Copyright (c) 2026 Anton Soloviev

Permission is hereby granted, free of charge, to any person obtaining a copy
of this software and associated documentation files (the "Software"), to deal
in the Software without restriction, including without limitation the rights
to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
copies of the Software, and to permit persons to whom the Software is
furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all
copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
SOFTWARE.

Copyright 2022 The Noto Project Authors (https://github.com/notofonts/latin-greek-cyrillic)

This Font Software is licensed under the SIL Open Font License, Version 1.1.
This license is copied below, and is also available with a FAQ at:
https://scripts.sil.org/OFL


-----------------------------------------------------------
SIL OPEN FONT LICENSE Version 1.1 - 26 February 2007
-----------------------------------------------------------

PREAMBLE
The goals of the Open Font License (OFL) are to stimulate worldwide
development of collaborative font projects, to support the font creation
efforts of academic and linguistic communities, and to provide a free and
open framework in which fonts may be shared and improved in partnership
with others.

The OFL allows the licensed fonts to be used, studied, modified and
redistributed freely as long as they are not sold by themselves. The
fonts, including any derivative works, can be bundled, embedded,
redistributed and/or sold with any software provided that any reserved
```

names are not used by derivative works. The fonts and derivatives,
however, cannot be released under any other type of license. The
requirement for fonts to remain under this license does not apply
to any document created using the fonts or their derivatives.

DEFINITIONS
"Font Software" refers to the set of files released by the Copyright
Holder(s) under this license and clearly marked as such. This may
include source files, build scripts and documentation.

"Reserved Font Name" refers to any names specified as such after the
copyright statement(s).

"Original Version" refers to the collection of Font Software components as
distributed by the Copyright Holder(s).

"Modified Version" refers to any derivative made by adding to, deleting,
or substituting -- in part or in whole -- any of the components of the
Original Version, by changing formats or by porting the Font Software to a
new environment.

"Author" refers to any designer, engineer, programmer, technical
writer or other person who contributed to the Font Software.

PERMISSION & CONDITIONS
Permission is hereby granted, free of charge, to any person obtaining
a copy of the Font Software, to use, study, copy, merge, embed, modify,
redistribute, and sell modified and unmodified copies of the Font
Software, subject to the following conditions:

1) Neither the Font Software nor any of its individual components,
in Original or Modified Versions, may be sold by itself.

2) Original or Modified Versions of the Font Software may be bundled,
redistributed and/or sold with any software, provided that each copy
contains the above copyright notice and this license. These can be
included either as stand-alone text files, human-readable headers or
in the appropriate machine-readable metadata fields within text or
binary files as long as those fields can be easily viewed by the user.

3) No Modified Version of the Font Software may use the Reserved Font
Name(s) unless explicit written permission is granted by the corresponding
Copyright Holder. This restriction only applies to the primary font name as
presented to the users.

4) The name(s) of the Copyright Holder(s) or the Author(s) of the Font
Software shall not be used to promote, endorse or advertise any
Modified Version, except to acknowledge the contribution(s) of the
Copyright Holder(s) and the Author(s) or with their explicit written
permission.

5) The Font Software, modified or unmodified, in part or in whole,
must be distributed entirely under this license, and must not be
distributed under any other license. The requirement for fonts to
remain under this license does not apply to any document created
using the Font Software.

TERMINATION
This license becomes null and void if any of the above conditions are

not met.

DISCLAIMER
THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL THE COPYRIGHT HOLDER BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE FONT SOFTWARE.